

37 Soil and Plant Nutrition

KEY CONCEPTS

- 37.1** Soil contains a living, complex ecosystem *p. 806*
- 37.2** Plant roots absorb many types of essential elements from the soil *p. 809*
- 37.3** Plant nutrition often involves relationships with other organisms *p. 812*

Study Tip

Use mnemonics: Mnemonics are memory tools to help remember facts. For example, “See Hopkins, California—Mighty good” (CHOPKNSCaMg) is a mnemonic for remembering the nine plant macronutrients, as shown below. Make your own mnemonics for remembering some functions of plant macronutrients (see Table 37.1).

See Hopkins, California—Mighty good

See = C = carbon
H = H = hydrogen
o = O = oxygen

➔ Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 37
- Animation: How Plants Obtain Minerals from Soil
- Figure 37.11 Walkthrough: The Roles of Soil Bacteria in the Nitrogen Nutrition of Plants

For Instructors to Assign (in Item Library)

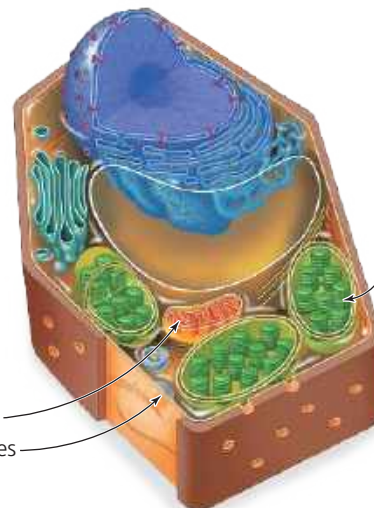
- Scientific Skills Exercise: Making Observations
- Tutorial: Nitrogen Nutrition in Plants



Figure 37.1 This farmer in India is applying chemical fertilizers to a field of rice. Nitrogen (N), phosphorous (P), and potassium (K) are commonly depleted from soils to such an extent as to limit crop productivity. That is why they are the major ingredients of most chemical fertilizers.

Why do plants need minerals from the soil?

Nitrogen, phosphorus, potassium, and other minerals are essential for plant growth because of their roles in the structure and function of plant cells.



Phosphorus is a component of

- DNA and RNA (throughout the cell)
- ATP produced by mitochondria
- phospholipids in cell membranes

Nitrogen is a component of

- DNA and RNA (throughout the cell)
- proteins (throughout the cell)
- chlorophyll

Potassium is a cofactor for enzymes used throughout the cell; plays a major role in maintaining turgor